

Coupling atoms to nanophotonic cavities for quantum network applications

EE6502 Course Presentation
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Enabling Quantum Networks

- Quantum network: Connect storage/processing nodes using q. comm. Channels to enable transmission of quantum information
- To realize a quantum network, is required: a traveling qubit to transmit the information, memory (storage) qubits to store/process information, **an interface** that can map storage qubits to traveling qubits.
- Strong contenders: Optical Photons for traveling (fast, low loss)
Atomic qubits for storage/QIP (long coherence times)

Enabling Quantum Networks

- How to interface the two?
- Leverage atom-photon interaction (eg: dipole transitions, ...)
- Fundamental problem: in free-space,
 - Atom's interaction cross-section: $\sigma \propto \lambda^2$ (λ : wavelength of transition $\sim 0.1\mu\text{m}$)
 - Beam cross-section: $A \propto w^2$ (w : beam cross-section $> 1\mu\text{m}$)

For deterministic absorption, $\sigma \gg A$

Enabling Quantum Networks

So how do we enhance σ ?

Possible solutions:

1. Use an ensemble of atoms
2. Use Rydberg excitations (much higher σ)
3. Enhance light-matter coupling using resonators (eg: cavities)

3 is the most straightforward of all

Atom in a cavity

What happens to atom-photon interaction in a cavity?

Photon repeatedly passes through the atom, enhanced absorption probability

→ σ enhanced by “finesse” of the cavity, F [3]

F : estimate of how number of bounces of the photon inside the cavity

Condition for enhanced coupling: $\sigma \times F \gg A$

Atom in a cavity

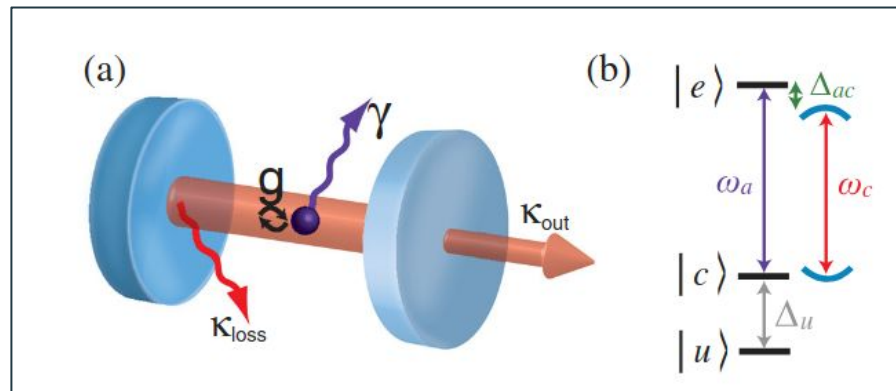
These parameters can be rewritten in terms of loss parameters, namely,

Using these, we define a parameter, **Cooperativity (C)**:

$$C = \frac{g^2}{2\kappa\gamma} \gg 1$$

where, g is the rabi frequency of the transition

$$\kappa = \frac{\pi C}{2L\mathcal{F}} \quad \gamma = \frac{\mu_{ce}^2 \omega^3}{6\pi\epsilon_0 \hbar c^3}$$



- a) Describing lossy processes for atom in a cavity
- b) Level structure diagram for a 3 level system; ω_c is the cavity mode, tuned to be close to the atomic transition ω_a [2]

Atom-photon interaction in a lossy cavity

In absence of losses, the interaction is given using the Jaynes-Cumming model:

Due to losses, there exist different regimes of coupling depending on g , k , γ

$$\mathcal{H}_{JC} = \underbrace{\hbar\omega_a\sigma_{ce}^\dagger\sigma_{ce}}_{\text{bare atom}} + \underbrace{\hbar\omega_c a^\dagger a}_{\text{empty cavity}} + \underbrace{\hbar g(\sigma_{ec}a + \sigma_{ce}a^\dagger)}_{\text{dipole coupling}}$$

Within $C \gg 1$, we are interested in the “Purcell regime” : $k > g \gg \gamma$

Interpretation: loss rate from cavity, $k > g \Rightarrow$ It is possible for atom to emit into output cavity mode, while being projected back to ground state (hence can time the next absorption cycle)

Recap...

- For a quantum network node, we need a way to interface flying (photonic) qubits with memory (atomic) qubits
- Cavities help realise deterministic atom-light interaction with $C = \frac{g^2}{2\kappa\gamma}$ (Cooperativity) as a figure of merit
- Atom can interact with cavity mode through Jaynes-Cumming interaction; we are interested in the regime where atom is able to emit into the output mode *deterministically*

Nanophotonic structures

- High cooperativity $\Rightarrow$ more deterministic absorption!
- Now, $g \propto \frac{1}{\sqrt{V}}$ $\Rightarrow$ A “nanophotonic cavity” allows for very high g (and hence C)
- From a neutral atom qc perspective, atoms are isolated and trapped in optical tweezers. (separated at $O(\sim \mu\text{m})$)
- Thus, smaller (nano)structures can also help couple multiple of these atoms at once!

In our case, these structures are realized using Photonic Crystals.

Experimental Sequence

Experiment

Goal: Realize coupling between a nanophotonic device and an atom in an optical tweezer

Measurement: Determine single-atom cooperativity via reflection measurement.

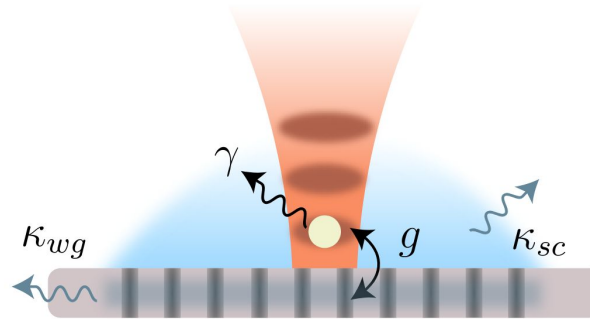


Fig: Atom in a tweezer coupling to evanescent modes of PC [1]

Control at Atomic level

$$\text{Atom speed at 300 K} \sim \sqrt{\frac{k_B T}{m}} \sim 2500 \text{ m/s}$$

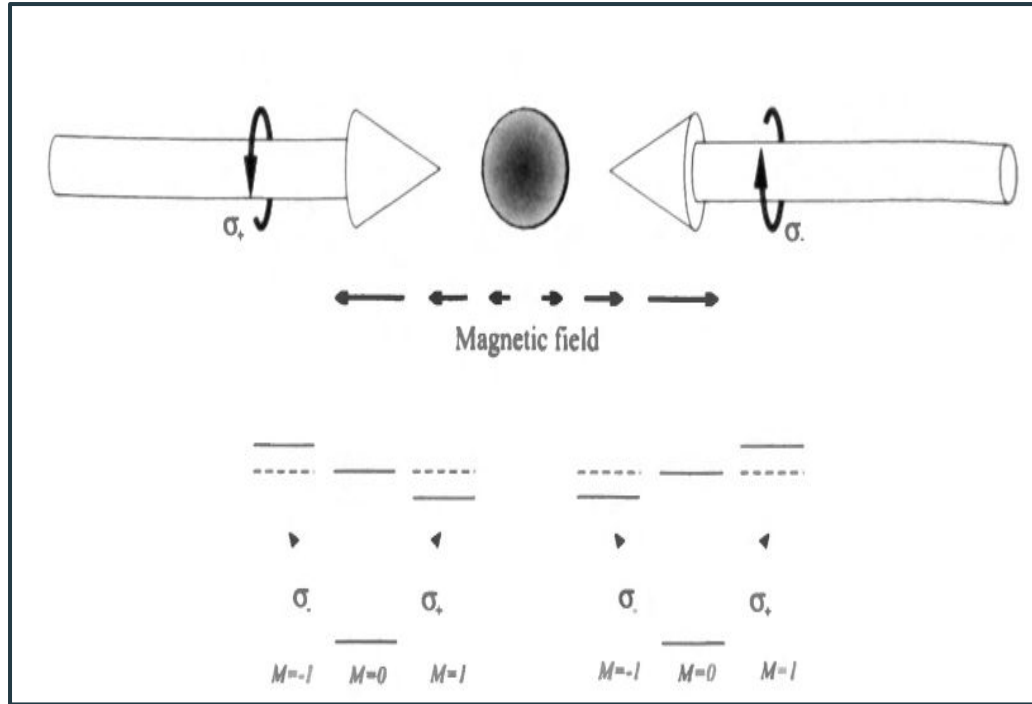
Need to **1. Cool the atoms**

2. Trap them

to be able to control at atomic level

Stage 1: Magneto-Optical Trap

“Laser Cooling” - slows down the atoms using momentum kicks



“Red detuned” beam to target

atoms using Doppler effect

$$\omega' = \omega \left(1 + \frac{v}{c} \right).$$

$$F = -\gamma v \longrightarrow \text{Slowing Force}$$

Inhomogeneous B field about Trap

center to trap the atom

$$F = -\gamma v - kx$$

Damped SHO => Forms a trap

Stage 2: Polarization Gradient Cooling

- Two-photon cooling process
- Creates a potential well around the atom due to polarization gradient; atom gets trapped near minima
- Cools the system all the way down to 15 μK (from 50 μK)

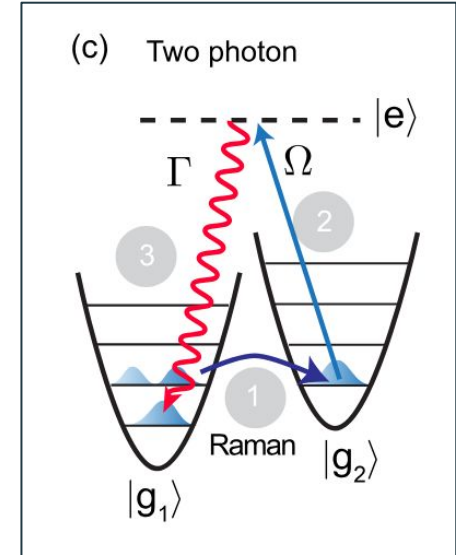


Fig: level diagram showing 2-photon processes leading to PGC [8]

Generating Optical Tweezers

- Uses a device called “Acousto-Optic Deflector”
- Contains an acoustic medium (eg TeO₂); when pumped with RF frequency f , it causes laser beam to diffract ($\Delta\theta \propto f$)
- Focused into a tight waist by High NA lens, leading to a deep potential well (hence can trap atoms!)

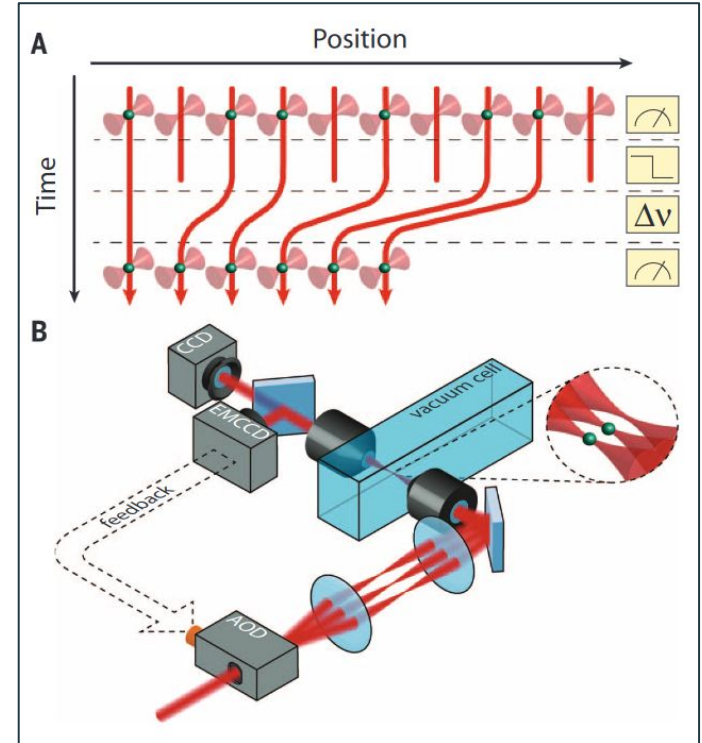
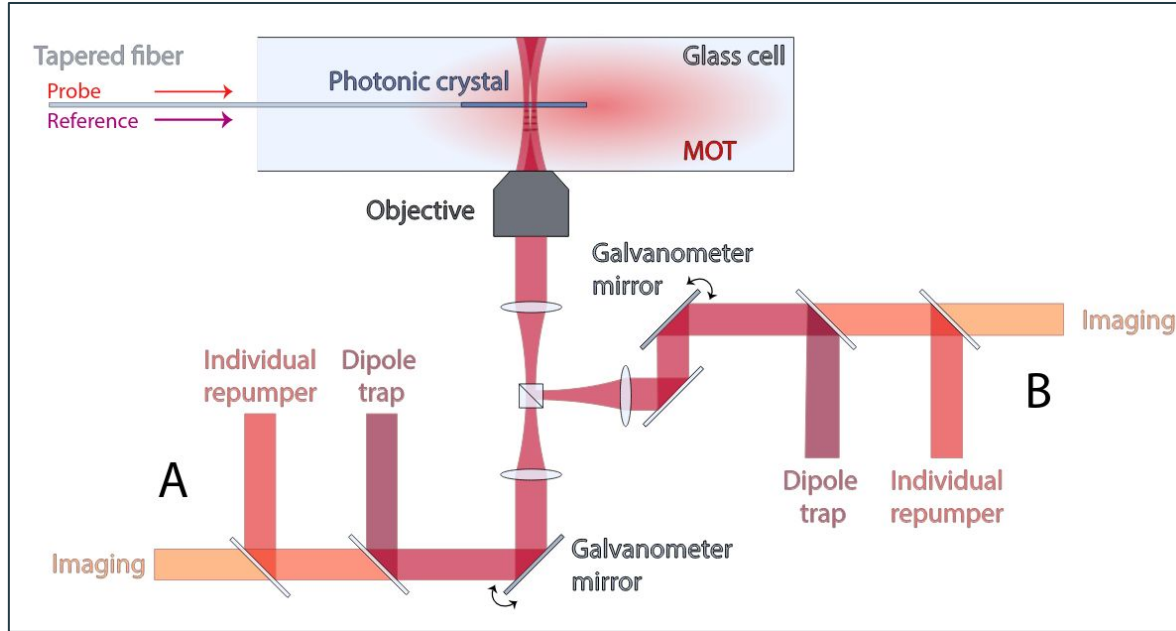


Fig: Example setup for preparing optical tweezers [7]

Experimental apparatus:

(780nm(Probe) , 795nm (Rep), 815nm (T))



Reflection off the glass cell collected for the purpose of spectroscopy

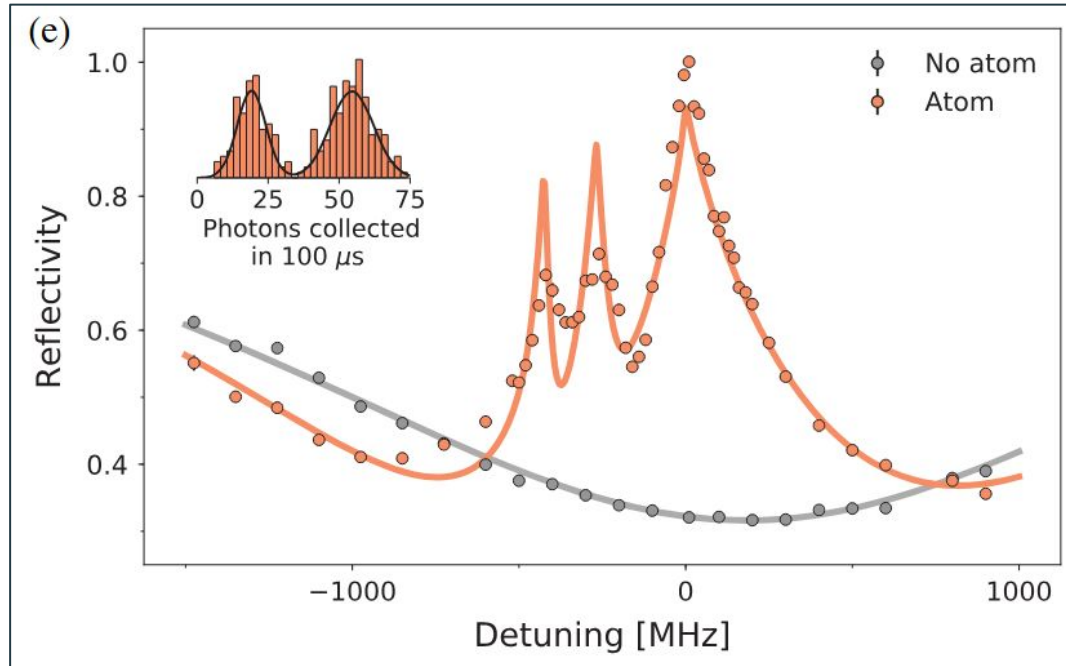
Probe beam sent through the Tapered fiber directly

Ref [1]

Experimental sequence [5]

1. Atoms undergo multiple stages of cooling (MOT, PGC) and then loaded into tweezer once cold enough
2. Using a galvo mirror, atoms are shuttled to location of the nanophotonic cavity
3. Atoms are located at sites $\sim 100\text{nm}$ outside the cavity (where evanescent fields exist); atoms thus couple to these evanescent modes of the PC
4. For reflection spectra, probe beam is scanned through a range of frequencies around atomic resonance ($5S_{1/2} F=2 \rightarrow 5P_{3/2} F' = 1,2,3$)

Results



We broadening of the lines due to “Purcell effect”

By fitting to a Cooperativity to reflectivity estimation model, the parameters are obtained as,

$$2g = 2\pi \times 1.24 \text{ GHz}$$

$$\Gamma = 2\pi \times 0.006 \text{ GHz}$$

$$k_{wg}, k_{sc} = 2\pi \times \{0.86, 2.77\} \text{ GHz}$$

$$\text{Cooperativity } C = 71$$

$$r = \kappa_{wg} \left(\frac{\kappa}{2} - i\delta_c + \sum_i \frac{g_i^2}{\gamma/2 - i\delta_i} \right)^{-1} - 1, \quad \text{Refs: [1], [5]}$$

Phys. Rev. Lett. **124**, 063602 (supp)

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References

- [1] Samutraphoot et al, Phys. Rev. Lett. 124, 063602
- [2] Reiserer, Rempe, Rev. Mod. Phys. 87, 1379
- [3] [Ryou Albert dissertation](#)
- [4] Hannes Bernien | Quantum networks with neutral atom processing nodes (MCQST)
- [5] [Ocola dissertation](#)
- [6] Dmitry Budker - Atomic Physics
- [7] Manuel Endres et al., Science 354, 1024-1027 (2016)
- [8] S. Phatak et al, Phys. Rev. A 110, 043116

Cooperativity estimates (theoretical)

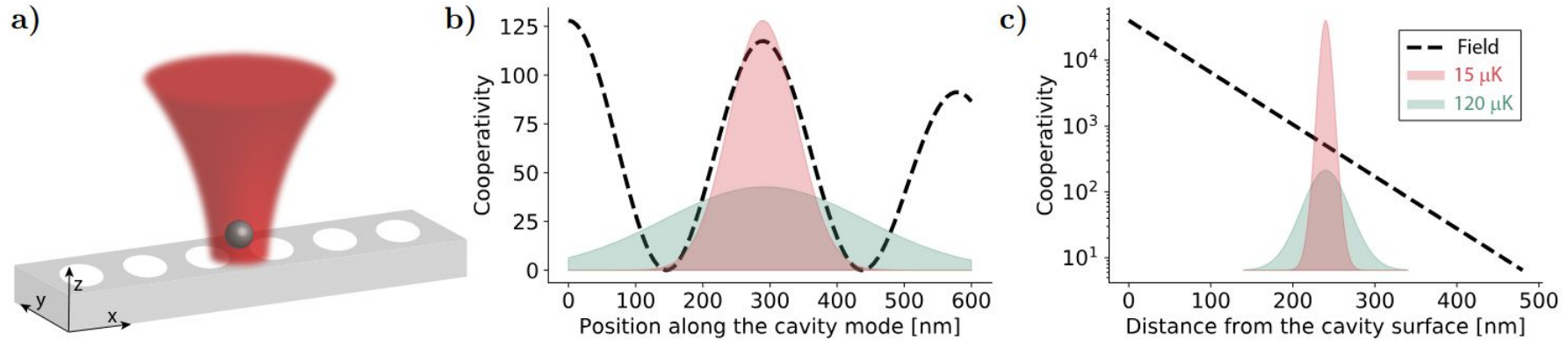


FIG. S6: **a)** Illustration of an atom trapped in a tweezer above the photonic crystal and definition of axes. **b)** Cooperativity variation in the longitudinal (x-axis) direction. The green and red dashed lines correspond to the atomic spatial width at 120 μK and 15 μK , respectively. **c)** Cooperativity variation and atomic spatial widths in the z direction.